

1. Administrative & Study Identification

Official Study Title: A Randomized, Open-label, Phase 3 Study of MK-2870 vs Chemotherapy (Docetaxel or Pemetrexed) in Previously Treated Advanced or Metastatic Nonsquamous Non-small Cell Lung Cancer (NSCLC) With EGFR Mutations or Other Genomic Alterations
Study Title: Sacituzumab Tirumotecan (MK-2870) Versus Chemotherapy in Previously Treated Advanced or Metastatic Nonsquamous Non-small Cell Lung Cancer (NSCLC) With EGFR Mutations or Other Genomic Alterations (MK-2870-004) **Short Title/ Acronym:** MK-2870-004 **Protocol Number:** 2870-004 **NCT Number:** NCT06074588 **Posting ID:** 412610573360339005 **Trial Status:** RECRUITING **Sponsor Name:** Merck **Company:** Merck **Investigational Product:** Sacituzumab tirumotecan (MK-2870) **Phase of Development:** Phase 3 **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether sacituzumab tirumotecan is superior to chemotherapy with respect to progression-free survival (PFS) and overall survival (OS) in NSCLC participants with EGFR mutations. **Secondary Objectives:** To evaluate PFS and OS in all participants with NSCLC, and to evaluate Objective Response Rate (ORR) of participants with NSCLC with EGFR Mutations.

2.2 Background & Rationale To address the critical unmet medical need for effective targeted therapies in patients with advanced or metastatic nonsquamous NSCLC with exon 19del or exon 21 L858R EGFR mutations or other genomic alterations (ALK gene rearrangements, ROS1 rearrangements, BRAF V600E mutations, NTRK gene fusions, MET exon 14 skipping mutations, RET rearrangements, or less common EGFR point mutations of exon 20 S768I, exon 21 L861Q, or exon 18 G719X). Trophoblast cell-surface antigen 2 (TROP2) is broadly expressed in NSCLC, and sacituzumab tirumotecan (a TROP2-directed antibody-drug conjugate) aims to offer improved clinical outcomes over standard chemotherapy options.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Docetaxel or Pemetrexed) **Blinding:** Open-label **Randomization:** 1:1

3.2 Planned Enrollment & Duration Trial Status: RECRUITING
Targeted Enrollment: 556 evaluable patients **Number of Sites:**
Global, Multicenter **Estimated Study Start:** 12-Nov-2023 **Estimated
Primary Completion:** 10-May-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Histologically- or cytologically-documented advanced (Stage III not eligible for resection or curative radiation) or metastatic non-squamous NSCLC with specific mutations (as defined in rationale). 3. Documentation of locally assessed radiological disease progression while on or after last treatment based on RECIST 1.1, with measurable disease per RECIST 1.1. 4. Participants with genome mutations must have received 1 or 2 prior lines of epidermal growth factor receptor tyrosine kinase inhibitor (EGFR TKI), including a third generation TKI for participants with a T790M mutation; and 1 platinum-based therapy after progression on or after EGFR TKI. 5. Archival tumor tissue sample or newly obtained biopsy of a tumor lesion not previously irradiated has been provided. 6. ECOG performance status of 0 or 1. 7. Participants with controlled HIV or Hepatitis B/C may be eligible per protocol specifications.

4.2 Exclusion Criteria 1. Predominantly squamous cell histology NSCLC or mixed tumor(s) with small cell elements. 2. Known active central nervous system (CNS) metastases and/or carcinomatous meningitis (stable, previously treated brain metastases are allowed). 3. Prior treatment with a TROP2-targeted antibody-drug conjugate (ADC) or a topoisomerase I-containing ADC. 4. Grade ≥ 2 peripheral neuropathy. 5. Documented severe dry eye syndrome, severe Meibomian gland disease, blepharitis, or corneal disease. 6. Active inflammatory bowel disease requiring immunosuppressive medication or previous history of inflammatory bowel disease. 7. Uncontrolled, significant cardiovascular disease or cerebrovascular disease. 8. Received a live or live-attenuated vaccine within 30 days before the first dose of study intervention.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Sacituzumab tirumotecan via intravenous (IV) infusion on Days 1, 15, and 29 of each 6-week treatment cycle. The control group will receive Docetaxel or Pemetrexed on Days 1 and 22 of each 6-week cycle.
Route of Administration: Intravenous (IV). **Duration of Treatment:** Until disease progression, unacceptable toxicity, or study withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Standard premedication for ADCs (e.g., diphenhydramine, H2 antagonist,

acetaminophen, dexamethasone). **Prohibited:** Concurrent systemic anticancer therapy; live or live-attenuated vaccines within 30 days prior to the first dose.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Sample Collection, Baseline Labs
Baseline	Day 1	Randomization, First Dose
Interim	Every 6 Weeks	Treatment cycles, Safety Labs, Tumor Imaging (every 6 weeks for first 54 weeks, then every 9 weeks)
End of Study	+30 Days Post-Dose	Safety follow-up, followed by long-term survival status follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- * **Progression-free Survival (PFS):** Assessed in participants with NSCLC with EGFR Mutations. Defined as the time from randomization to the first documented disease progression per RECIST 1.1 based on blinded independent central review (BICR) or death due to any cause.
- * **Overall Survival (OS):** Assessed in participants with NSCLC with EGFR mutations. Defined as the time from randomization to death due to any cause.

7.2 Secondary Endpoints

- * **[Endpoint 1]** PFS of All Participants with NSCLC.
- * **[Endpoint 2]** OS of All Participants with NSCLC.
- * **[Endpoint 3]** Objective Response Rate (ORR) of Participants with NSCLC with EGFR Mutations.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous clinical and laboratory monitoring graded via CTCAE during treatment and up to 30 days post-treatment. **Serious Adverse Events (SAEs):** Reported within 24 hours of investigator awareness to the sponsor. **Data Monitoring Committee (DMC):** External independent Data Monitoring Committee assigned by the sponsor to evaluate trial safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy evaluation; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** \$N = 556\$ adequately powered to demonstrate superiority in PFS and OS primarily within the EGFR-mutated cohort. **Handling of Missing Data:** Censoring applied per standard RECIST 1.1 guidelines for time-to-event endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite visits and centralized remote data monitoring by the sponsor. **Audit Trail:** Validated Electronic Case Report Forms (eCRF) leveraging automated edit checks, query tracking, and comprehensive audit trails.

11. Study Identifiers & Governance

Primary ID: 2870-004 **Registry Number:** NCT06074588 **Posting ID:** 412610573360339005 **Sponsor/Lead Organization:** Merck **Company:** Merck **Study Title:** Sacituzumab Tirumotecan (MK-2870) Versus Chemotherapy in Previously Treated Advanced or Metastatic Nonsquamous Non-small Cell Lung Cancer (NSCLC) With EGFR Mutations or Other Genomic Alterations (MK-2870-004) **Official Regulatory Title:** A Randomized, Open-label, Phase 3 Study of MK-2870 vs Chemotherapy (Docetaxel or Pemetrexed) in Previously Treated Advanced or Metastatic Nonsquamous Non-small Cell Lung Cancer (NSCLC) With EGFR Mutations or Other Genomic Alterations

12. Clinical Framework (NSCLC Specific)

Primary Condition: Non-small Cell Lung Cancer (NSCLC) **Disease Area:** Non-small Cell Lung Cancer (NSCLC) **Preferred UMLS Name:** Non-small Cell Lung Carcinoma **Diagnosis Threshold:** Advanced or metastatic nonsquamous NSCLC with EGFR mutations or other actionable genomic alterations **Medical Methodology:** TROP2-directed Antibody-Drug Conjugate (ADC) vs. Standard Chemotherapy **Optimization Target:** Progression-Free Survival (PFS) and Overall Survival (OS) extension

13. Study Procedures & Methodology

Management Pathway: Systemic anti-cancer therapy clinical pathway for advanced NSCLC **Education Protocol (HCP):** Site initiation visits, rigorous protocol instruction, and RECIST 1.1 training **Education Protocol (Patient):** Informed consent process, treatment schedule, and side-effect management counseling **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for tumor imaging and source data verification (SDV)

14. Observation & Follow-Up Schedule

Total Duration: Estimated until March 2030 (Study Completion) **Onsite Visit Cadence:** Every 3 to 6 weeks aligned with systemic treatment cycles **Remote Monitoring Frequency:** Continuous AE/SAE

monitoring and periodic survival follow-up calls after treatment discontinuation **Checklist Review Interval:** Radiographic tumor imaging assessments every 6 weeks for 54 weeks, then every 9 weeks

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	06-Apr-2026				
Clinical Operations Lead	[Name]	_____	06-Apr-2026		Lead		
Statistician	[Name]	_____	06-Apr-2026				